2. Matrix Algebra

2.1 Matrix Operations

- **Main Diagonal**: The diagonal entries in matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, \ldots, a_{nn}$. They form the main diagonal of A.
- **Diagonal Matrix**: A diagonal matrix is a square $n \times n$ matrix whose non-diagonal entries are zero.

Theorem 1

Let A, B, C be matrices of the same size, and let r and s be scalars.
a. $A + B = B + A$
b. $(A + B) + C = A + (B + C)$
c. $A + 0 = A$
d. $r(A + B) = rA + rB$
e. $(r + s)A = rA + sA$
f. $r(sA) = (rs)A$

Definition (Matrix Multiplication)

If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $\vec{b}_1, \vec{b}_2, \ldots, \vec{b}_p$, then the product AB is the $m \times p$ matrix whose columns are $A\vec{b}_1, A\vec{b}_2, \ldots, A\vec{b}_p$. That is:

$$AB = A \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \cdots & \vec{b}_p \end{bmatrix} = \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_p \end{bmatrix}$$

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B.

Row-Column Rule for Computing AB

If the product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B.
If $(AB)_{ij}$ denotes the (i, j)-entry in AB, and if A is an $m \times n$ matrix, then:
$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$

- $row_i(AB) = row_i(A) \cdot B$
- $column_j(AB) = A \cdot column_j(B)$

Theorem 2

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.
a. $A(BC) = (AB)C$
b. $A(B + C) = AB + AC$
c. $(B + C)A = BA + CA$
d. $r(AB) = (rA)B = A(rB)$
e. $I_m A = AI_n = A$

Warnings

1. In general, $AB \neq BA$
2. The cancellation law does not hold for matrix multiplication: $AB = AC$ does not imply $B = C$ (in general)
3. If a product AB is the zero matrix, we can't conclude that either $A = 0$ or $B = 0$ (in general)

Definition (Matrix Powers)

If A is an $n \times n$ matrix and if k is a positive integer, then:

$$A^k = \underbrace{A \cdot A \cdot \cdots \cdot A}_{k \text{ times}}$$

- $A^{km} = (A^k)^m$
- If $AB = BA$, then:
  - $(AB)^k = A^k B^k$
  - $(A \pm B)^2 = A^2 \pm 2AB + B^2$
  - $(A - B)(A + B) = A^2 - B^2$

Theorem 3 (Transpose Properties)

a. $(A^\top)^\top = A$
b. $(A + B)^\top = A^\top + B^\top$
c. $(rA)^\top = rA^\top$
d. $(AB)^\top = B^\top A^\top$

The transpose of a product of matrices equals the product of their transposes in the reverse order.

2.2 The Inverse of a Matrix

- **Invertible Matrix**: An $n \times n$ matrix A is invertible if there is an $n \times n$ matrix A^{-1} so that $A^{-1}A = I_n$ and $AA^{-1} = I_n$. A^{-1} is uniquely determined by A.
- **Singular Matrix**: A matrix that is not invertible.
- **Nonsingular Matrix**: A matrix that is invertible.

Theorem 4

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- The quantity $ad - bc$ is the determinant of A, denoted $\det A = ad - bc$.
- A 2×2 matrix is invertible if and only if $\det A \neq 0$. If $ad - bc = 0$, then A is not invertible.

Theorem 5

If A is invertible, then the equation $Ax = b$ has the unique solution $x = A^{-1}b$.

Theorem 6

a. If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$
b. If A and B are invertible $n \times n$ matrices, then so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$
c. $(A^\top)^{-1} = (A^{-1})^\top$
d. If λ is a nonzero scalar, then $(\lambda A)^{-1} = \lambda^{-1}A^{-1} = \frac{1}{\lambda}A^{-1}$

The power of a matrix

A is an invertible $n \times n$ matrix and k is a nonnegative integer, then

1. $A^{-k} = (A^{-1})^k$
2. $(A^\lambda)^\mu = A^{\lambda\mu}$
3. $A^0 = 0$
4. $A^\lambda A^\mu = A^{\lambda+\mu}$

The product of nonsingular (invertible) matrices is invertible, and the inverse is the product of their inverses in the reverse order:

$$(A_1 A_2 \cdots A_p)^{-1} = A_p^{-1} \cdots A_2^{-1} A_1^{-1}$$

Elementary Matrices

- If an elementary row operation is performed on an $m \times n$ matrix A, the resulting matrix can be written as EA, where the $m \times m$ matrix E is created by performing the same row operation on I_m.
- Each elementary matrix E is invertible.
- The inverse of E is the elementary matrix of the same type that transforms E back into I.

Theorem 7

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n, and any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1}.

Algorithm for Finding A^{-1}

1. Row reduce the augmented matrix $[A \mid I_n]$
2. If A is row equivalent to I_n, then $[A \mid I_n]$ row reduces to $[I_n \mid A^{-1}]$
3. Otherwise, A is not invertible.

2.3 Characterizations of Invertible Matrix

Theorem 8: The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent (all true or all false):
a. A is an invertible matrix
b. A is row equivalent to the $n \times n$ identity matrix I_n
c. A has n pivot positions
d. The equation $Ax = 0$ has only the trivial solution
e. The columns of A form a linearly independent set
f. The linear transformation $\vec{x} \to A\vec{x}$ is one-to-one
g. The equation $Ax = b$ has at least one solution for each $\vec{b}$ in $\mathbb{R}^n$

h. The columns of A span $\mathbb{R}^n$
i. The linear transformation $\vec{x} \to A\vec{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$
j. There is an $n \times n$ matrix C so that $CA = I_n$
k. There is an $n \times n$ matrix D so that $AD = I_n$
l. $A^\top$ is an invertible matrix

Square matrices and its inverse

Let A and B be inverse matrices. If $AB = I$, the A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$

Theorem 9 (Invertible Linear Transformations)

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T. Then T is invertible if and only if A is an invertible matrix. The linear transformation S given by $S(\vec{x}) = A^{-1}\vec{x}$ is the unique function satisfying:

- $S(T(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$
- $T(S(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$

Definition

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is invertible if there exists a function S such that $S(T(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$ and $T(S(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$. We call S the inverse of T, and write it as T^{-1}.

2.4 Partitioned Matrices

Example 1: Partitioned Matrix Representation

The matrix

$$A = \begin{bmatrix} 3 & 0 & -1 & | & 5 & 9 & -2 \\ -5 & 2 & 4 & | & 0 & -3 & 1 \\ -8 & -6 & 3 & | & 1 & 7 & -4 \end{bmatrix}$$

can also be written as a 2×3 **partitioned (or block) matrix**:

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

Transpose of a Partitioned Matrix

If A is a partitioned matrix:

$$A = \begin{bmatrix} A_{11} & \dots & A_{1r} \\ \vdots & \ddots & \vdots \\ A_{s1} & \dots & A_{sr} \end{bmatrix}$$

then its transpose A^T is:

$$A^T = \begin{bmatrix} A_{11}^T & \dots & A_{s1}^T \\ \vdots & \ddots & \vdots \\ A_{1r}^T & \dots & A_{sr}^T \end{bmatrix}$$

Theorem 10: Column-Row Expansion of AB

If A is $m \times n$ and B is $n \times p$, then:

$$AB = \begin{bmatrix} \text{col}_1(A) & \text{col}_2(A) & \dots & \text{col}_n(A) \end{bmatrix} \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix}$$

$$= \text{col}_1(A)\text{row}_1(B) + \text{col}_2(A)\text{row}_2(B) + \dots + \text{col}_n(A)\text{row}_n(B)$$

Partitioned A (by rows) & B (by columns)

Let $A = (a_{ij})_{m \times s}$ (partitioned by rows):

$$A = \begin{bmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix}$$

Let $B = (b_{ij})_{s \times n}$ (partitioned by columns):

$$B = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \dots & \vec{b}_n \end{bmatrix}$$

Then:

$$AB = \begin{bmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix} \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \dots & \vec{b}_n \end{bmatrix} = \begin{bmatrix} \vec{a}_1^T \vec{b}_1 & \vec{a}_1^T \vec{b}_2 & \dots & \vec{a}_1^T \vec{b}_n \\ \vec{a}_2^T \vec{b}_1 & \vec{a}_2^T \vec{b}_2 & \dots & \vec{a}_2^T \vec{b}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a}_m^T \vec{b}_1 & \vec{a}_m^T \vec{b}_2 & \dots & \vec{a}_m^T \vec{b}_n \end{bmatrix}$$

Block Diagonal Matrix

A **block diagonal matrix** is a partitioned matrix with zero blocks off the main diagonal (of blocks).

Such a matrix is invertible if and only if each block on the diagonal is invertible.

2.5 Matrix Factorizations

Definition of LU Factorization

Assume that A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges. Then A can be written in the form $A = LU$, where:

- L is an $m \times m$ **lower triangular matrix** with 1's on the diagonal (called a **unit lower triangular matrix**, and it is invertible),
- U is an $m \times n$ echelon form of A.

This decomposition is called an **LU factorization** of A.

Solving $A\vec{x} = \vec{b}$ via LU Factorization

When $A = LU$, the equation $A\vec{x} = \vec{b}$ can be rewritten as $L(U\vec{x}) = \vec{b}$. Let $y = U\vec{x}$, then solve the pair of equations:

$$\begin{cases} L\vec{y} = \vec{b} \\ U\vec{x} = \vec{y} \end{cases}$$

Steps:

1. First solve $L\vec{y} = \vec{b}$ for $\vec{y}$:
   Row reduce the augmented matrix $[L\ \vec{b}] \to [I\ \vec{y}]$
2. Then solve $U\vec{x} = \vec{y}$ for $\vec{x}$:
   Row reduce the augmented matrix $[U\ \vec{y}] \to [I\ \vec{x}]$

Derivation via Elementary Matrices

Suppose A can be reduced to echelon form U using only **row replacements** (add a multiple of one row to another row below it). If there exist unit lower triangular elementary matrices $E_1, \ldots, E_p$ such that:

$$E_p \cdots E_1 A = U$$

then:

$$A = (E_p \cdots E_1)^{-1} U = LU$$

where:

$$L = (E_p \cdots E_1)^{-1} = E_1^{-1} \cdots E_p^{-1}$$

The row operations that reduce A to U will also reduce L to I.

Algorithm for LU Factorization

1. Reduce A to an echelon form U by a sequence of row replacement operations (if possible).
2. Place entries in L such that the same sequence of row operations reduces L to I.

Example 2: LU Factorization of a Matrix

Given matrix:

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Step 1: Reduce A to echelon form U

Row reduce A via row replacements:

$$A \sim A_1 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \sim A_2 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix} \sim U = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

Step 2: Construct L (unit lower triangular matrix)

From the row operations, L is:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

2.8 Subspaces of $\mathbb{R}^n$

Definition: Subspace

A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:
a. The zero vector is in H
b. For each $\vec{u}$ and $\vec{v}$ in H, the sum $\vec{u} + \vec{v}$ is in H (closed under addition)
c. For each $\vec{u}$ in H and each scalar c, the vector $c\vec{u}$ is in H (closed under scalar multiplication)

- The set $\mathrm{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\}$ is called the subspace spanned (or generated) by $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$.

Definition: Column Space

The column space of a matrix A, denoted $\mathrm{Col}\ A$, is the set of all linear combinations of the columns of A. The column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$.

Definition: Null Space

The null space of a matrix A, denoted $\mathrm{Nul}\ A$, is the set of all solutions of the homogeneous equation $A\vec{x} = 0$.

Theorem 12

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $A\vec{x} = 0$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Definition: Basis

A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H.

- The set $\{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n\}$ is the standard basis for $\mathbb{R}^n$.

2.9: Dimension and Rank

Definition: Coordinate Vector

Suppose the set $\mathcal{B} = \{\vec{b}_1, \ldots, \vec{b}_p\}$ is a basis for a subspace H.

For each $\vec{x}$ in H, the **coordinates of $\vec{x}$ relative to the basis $\mathcal{B}$** are the weights $c_1, \ldots, c_p$ such that:
$\vec{x} = c_1\vec{b}_1 + \cdots + c_p\vec{b}_p$

The vector in $\mathbb{R}^p$:

$$[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of $\vec{x}$ (relative to $\mathcal{B}$)** or the **$\mathcal{B}$-coordinate vector of $\vec{x}$**.

Definition: Dimension

The **dimension** of a nonzero subspace H, denoted by $\dim H$, is the number of vectors in any basis for H.

The dimension of the zero subspace is defined to be 0.

Definition: Rank

The **rank** of a matrix, denoted by $\text{rank } A$, is the dimension of the column space of A (i.e., $\text{rank } A = \dim \text{Col } A$).

Theorem 14: The Rank Theorem

If matrix A has n columns, then:

$$\text{rank } A + \dim \text{Null } A = n$$

Theorem 15: The Basis Theorem

Let H be a p-dimension subspace of $\mathbb{R}^n$:

- Any linearly independent set of exactly p elements in H is automatically a basis for H.
- Any set of p elements of H that spans H is automatically a basis for H.

Theorem 8: The Invertible Matrix Theorem (Continued)

Let A be an $n \times n$ matrix. The following statements are each equivalent to "A is invertible":

- m. The columns of A form a basis of $\mathbb{R}^n$.
- n. $\text{Col } A = \mathbb{R}^n$.
- o. $\dim \text{Col } A = n$.
- p. $\text{rank } A = n$.
- q. $\text{Null } A = \{\vec{0}\}$.
- r. $\dim \text{Null } A = 0$.